

VIGNANA BHARATHI INSTITUTE OF TECHNOLOGY

(Autonomous)

II B. Tech II Semester Supplementary End Examinations, June-2026**FLUID MECHANICS AND HYDRAULIC MACHINERY**

Time: 3 Hours

(ME)

Max. Marks: 70

PART-A 20 Marks

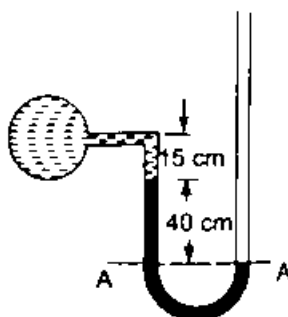
(Answer all questions)

1.
 - a. Define viscosity and mention its SI unit. 2 M
 - b. What is surface tension in fluids? 2 M
 - c. Define surface forces and body forces in fluid flow. 2 M
 - d. State Euler's equation for fluid flow. 2 M
 - e. Define turbo machinery. 2 M
 - f. What is meant by jet striking centrally on a vane? 2 M
 - g. What is Kaplan turbine? 2 M
 - h. Define water hammer. 2 M
 - i. What is delivery head in a pump? 2 M
 - j. What is an indicator diagram? 2 M

PART-B 50 Marks

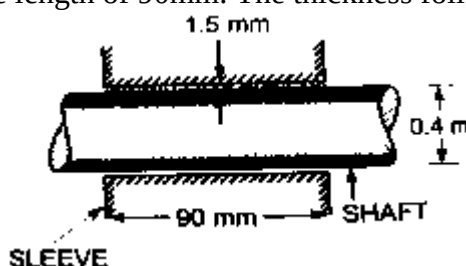
(Answer any one full question. Each question carries 10 marks)

2. A simple U – tube manometer containing mercury is connected to a pipe in which a fluid of sp.gr. 0.8 And having vacuum pressure is flowing. The other end of the manometer is open to atmosphere. Find the vacuum pressure in pipe, if the difference of mercury level in the two limbs is 40cm. and the height of the fluid in the left tube from the centre of pipe is 15cm below. 10 M



OR

3. The dynamic viscosity of oil used for lubrication between a shaft and sleeve is 6 poise. The shaft dia. is 0.4m and rotates at 190 rpm. Calculate the power lost in the bearing for a sleeve length of 90mm. The thickness foil film is 1.5mm. 10 M



4. A pitot tube is placed in the centre of a 300mm pipe line has one end pointing upstream and other perpendicular to it. The mean velocity in the pipe is 0.80 of the central velocity. Find the discharge through the pipe, if the pressure difference between the two orifices is 60mm of water. Co-efficient of Pitot tube $C_V = 0.98$. **10 M**

OR

5. Find the discharge of water flowing through a pipe 20 cm diameter placed in an inclined position, where a venturi-meter is inserted, having a throat diameter of 10 cm. The difference of pressure between the main and throat is measured by a liquid of specific gravity 0.4 in an inverted U-tube, which gives a reading of 30 cm. The loss of head between the main and throat is 0.2 times the kinetic head of pipe. **10 M**

6. A jet of water having a velocity of 15 m/s, strikes a curved vane which is moving with a velocity of 5 m/s in the same direction as that of the jet at inlet. The vane is so shaped that the jet is deflected through 135° . The diameter of jet is 100 mm. Assuming the vane to be smooth, find:
(i) Force exerted by the jet on the vane in the direction of motion,
(ii) Power exerted on the vane, and
(iii) Efficiency of the vane. **10 M**

OR

7. Describe the concept of pumped storage plants and explain the storage requirements of hydroelectric power stations with suitable examples. **10 M**

8. The external and internal diameters of an inward flow reaction turbine are 1.20 m and 0.6 m respectively. The head on the turbine is 22 m and velocity of flow through the runner is constant and equal to 2.5 m/s. The guide blade angle is given as 10° and the runner vanes are radial at inlet. If the discharge at outlet is radial, determine :
(i) The speed of the turbine,
(ii) The vane angle at outlet of the runner, and
(iii) Hydraulic efficiency. **10 M**

OR

9. A Francis turbine with an overall efficiency of 70% is required to produce 147.15 kW. It is working under a head of 8 m. The peripheral velocity $= 0.30 \sqrt{2gH}$ and the radial velocity of flow at inlet is $0.96 \sqrt{2gH}$. The wheel runs at 200 r.p.m. and the hydraulic losses in the turbine are 20% of the available energy. Assume radial discharge, determine: (i) The guide blade angle, (ii) The wheel vane angle at inlet, (iii) Diameter of the wheel at inlet, and (iv) Width of wheel at inlet. **10 M**

10. Explain the construction and working principle of a centrifugal pump. Derive the expression for work done and discuss various efficiencies of the pump. **10 M**

OR

11. Find the maximum speed of a single-acting reciprocating pump to avoid separation, which occurs at 3.0 m of water (abs.) The pump has a cylinder of diameter 10 cm and a stroke length of 20 cm. The pump draws water from a sump and delivers to a tank. The water level in the sump is 3.5 m below the pump axis and in the tank the water level is 13 m above the pump axis. The diameter and length of the suction pipe are 4 cm and 5 m while of delivery pipe the diameter and length are 3 cm, 20 m respectively. Take atmospheric pressure head $= 10.3$ m of water. **10 M**